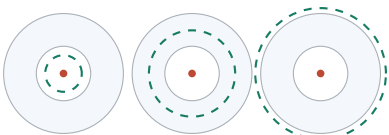
M3-010a 三个高斯面的包围电荷

红点为 Q；蓝色为介质球壳；绿圈为高斯面



$r < a$　　$a < r < b$　　$r > b$

从左到右：不包围束缚面、只包围内表面、包围两面

$$r < a\colon\ q_f = Q,\ q_b = 0$$

$$a < r < b\colon\ q_f = Q,\ q_b = -Q\alpha$$

$$r > b\colon\ q_f = Q,\ q_b = -Q\alpha + Q\alpha = 0$$